GATE-AII BRANCHES
Engineering Mathematics



PROBABILITY AND STATISTICS

Lecture No.- 18



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Recap of Previous Lecture



Topic

Concepts and questions based on poissons distribution

Topics to be Covered





Diwali

Topic — Exponential distribution — Basic

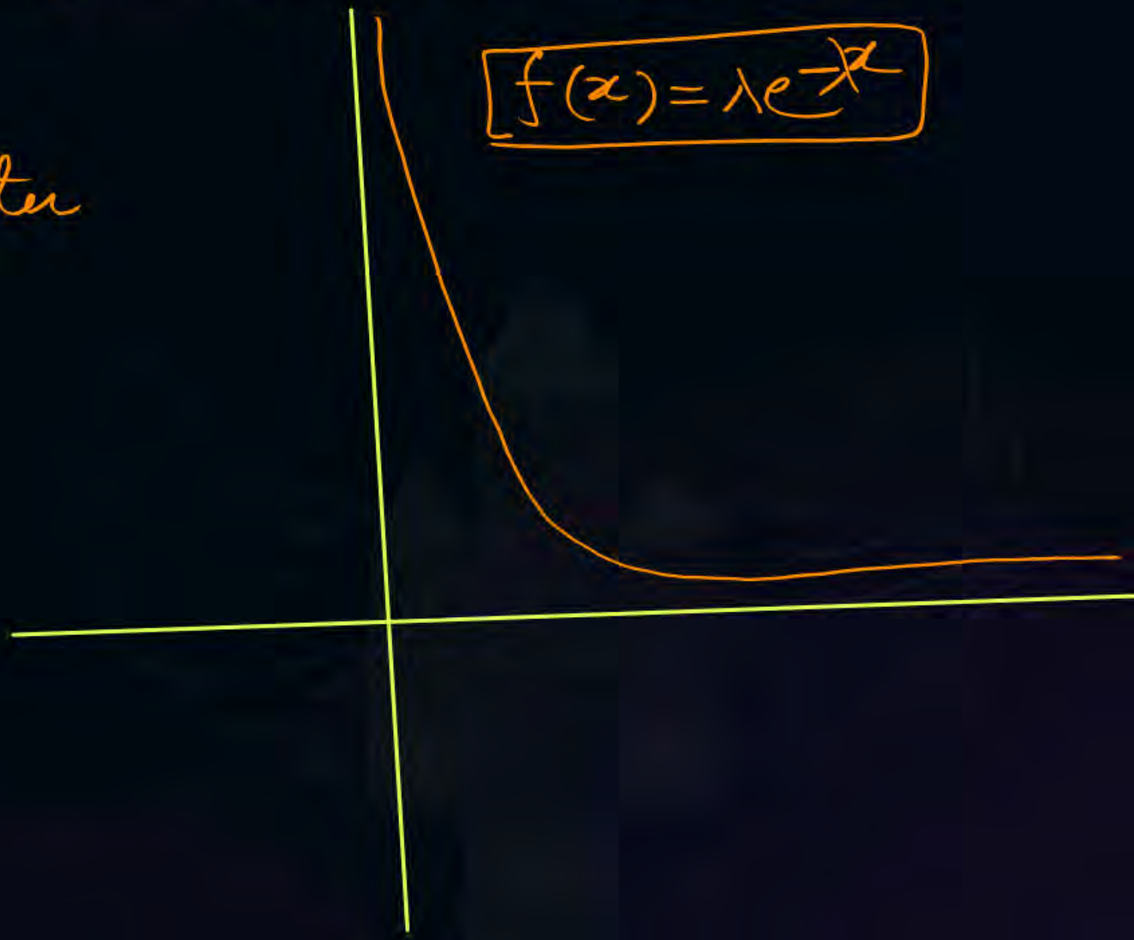
Topic — Questions based on exponential distribution

GATE - previous year - question

Exponential Distribution: Waiting time Distribution

Exponential Distribution — continuous Random var





in Poisson



Average No. of Person

small interval → time = continuous

If 1 min / 1 hrs. (H) Waiting Time

$$f(x) = \lambda e^{-\lambda x} \quad x > 0$$

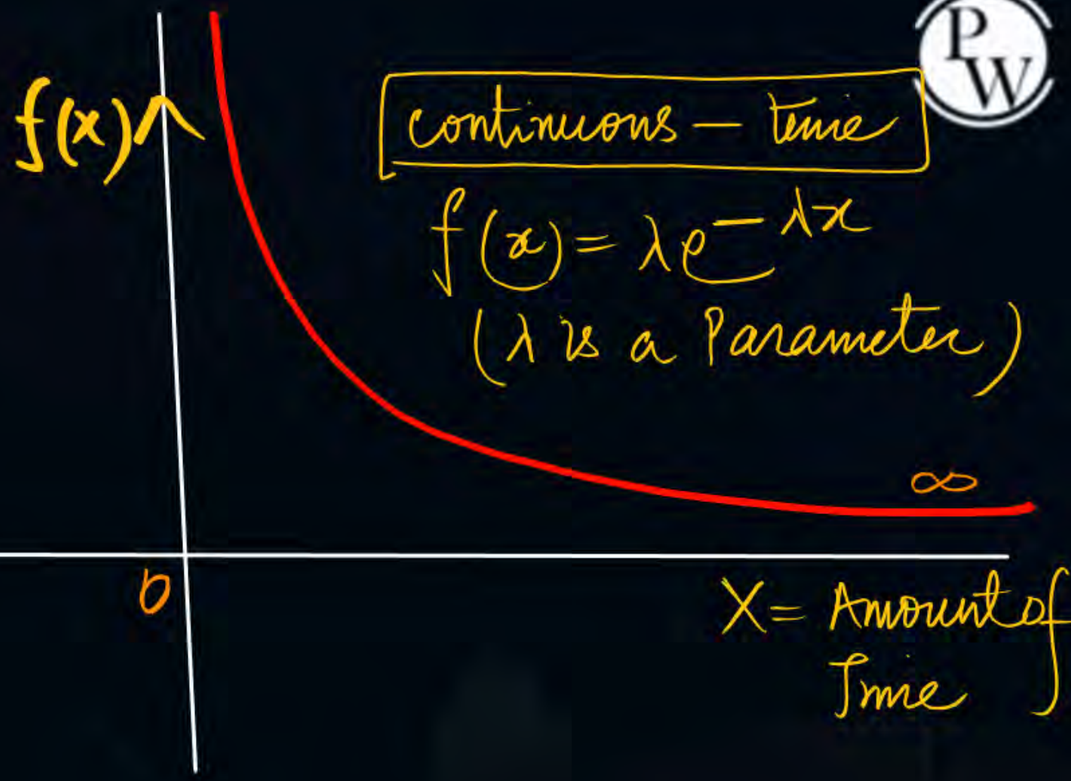
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$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{0}^{\infty} \lambda e^{-\lambda x}\,dx = \left[\lambda \frac{e^{-\lambda x}}{-\lambda}\right]_{0}^{\infty} = 1$$

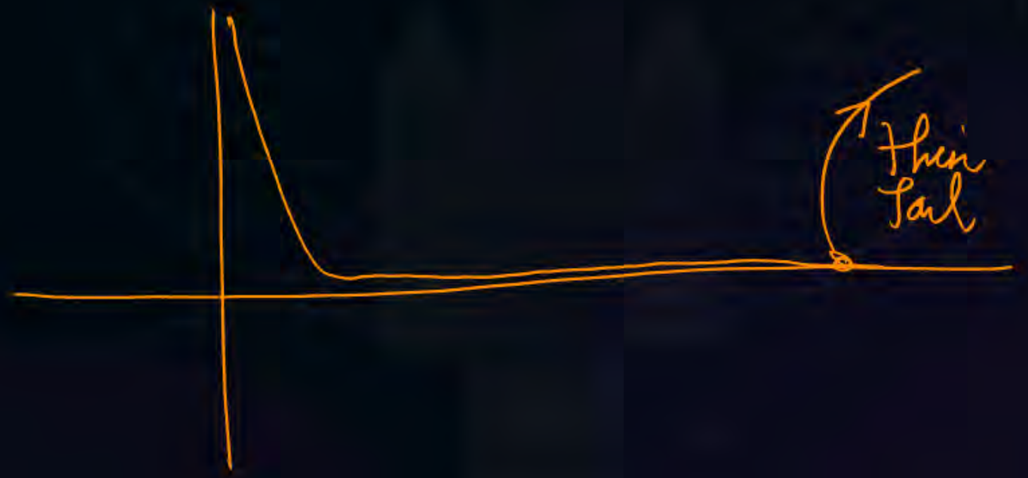
X = Amount of time

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$



Statistical Average

$$\text{mean } E[X] = \int_{0}^{\infty} x \lambda e^{-\lambda x}\,dx$$



Mean $E[X] = \int_0^\infty x \lambda e^{-\lambda x} dx = \frac{1}{\lambda}$

Mean $\exp(\lambda) = \frac{1}{\lambda}$

Variance $= E[X^2] - [E[X]]^2$

$V(X) = \int_0^\infty x^2 \cdot \lambda e^{-\lambda x} dx - \left[\int_0^\infty x e^{-\lambda x} dx\right]^2$

$$\boxed{V(X) = \frac{1}{\lambda^2}}$$

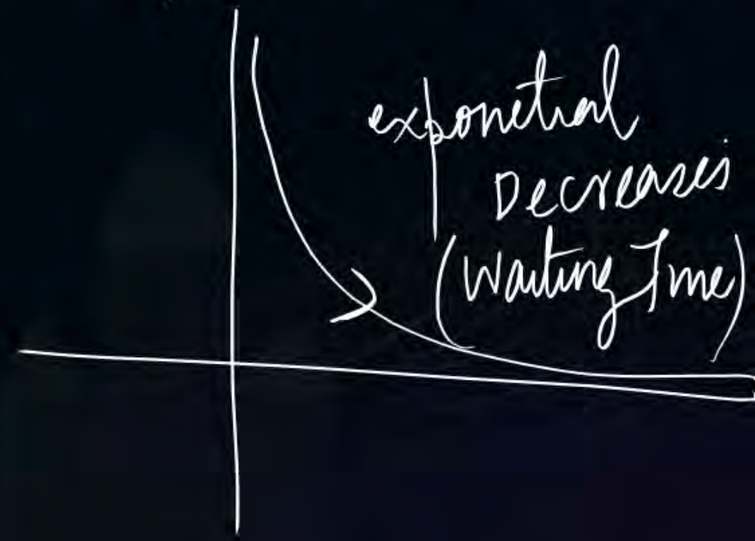
Standard deviation $= \sqrt{\text{variance}}$

$= \sqrt{\frac{1}{\lambda^2}}$

$= \frac{1}{\lambda}$

S.D

$f(x) = \lambda e^{-\lambda x} \quad x > 0$





Topic : Exponential distribution

#Q. What are the mean and variance of the exponential distirbution given by $f(x) = 3e^{-3x}$, $x \geq 0$

$$\text{mean} = \frac{1}{\lambda} = \frac{1}{3}$$

$$\text{variance} = \frac{1}{\lambda^2} = \frac{1}{3^2} = \frac{1}{9}$$

$f(x) = 3e^{-3x}$

$f(x) = \lambda e^{-\lambda x}$ $\lambda = 3$

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$



Topic : Exponential distribution

4 months

Statistical Averages — mean, V(x), S.D

#Q. Obtain the value of $k > 0$ for which the function given by $f(x) = 2e^{-kx}$, $x \geq 0$ follows an exponential distribution.

$f(x) = 2e^{-kx}$ compare with $\exp(\lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$

$k = 2$



Topic : Exponential distribution

$exp(\lambda) = \lambda e^{-\lambda x}$

$exp(\lambda) = exp(\frac{1}{2})$

#Q. An institute purchases laptop from either vendor V_1 or V_2 with equal proabability. Then lifetimes (in years) of laptop from vendor V_1 have a $U(0, 4)$ distribution, and the lifetime (in years) of laptop from vendor V_2 have an $Exp(1/2)$ distribution. If a randomly selected laptop in the institute has lifetime more than two years, then the probability that it was suppose by vendor V_2 is

Uniform Distributed

A $\frac{2}{2+e}$

B $\frac{1}{1+e}$

C $\frac{1}{1+e^{-1}}$

D $\frac{2}{2+e^{-1}}$

Uniform Distributed $U[0,4]$



$$f(x) = \frac{1}{(b-a)} = \frac{1}{(4-0)} = \frac{1}{4}$$

$$P(X \geq 2) = \int_2^4 \frac{1}{4}\, dx$$

$$= \frac{1}{4}(4-2) = \frac{2}{4} = \frac{1}{2}$$

START

$P(E_1) = \frac{1}{2}$ V_1 vendor

$P(E_2) = \frac{1}{2}$ for Path V_2 (vender)

$P\left(\frac{A}{E_1}\right) = \frac{1}{2}$

e^{-1}

More Than 2 YEARS
A

$\exp\left(\frac{1}{2}\right)$ $\lambda = \frac{1}{2}$ $P(X \geq 2) = e^{-\lambda x}$

$= e^{-\frac{1}{2} \times 2} = e^{-1}$

$P(X \geq x) = e^{-\lambda x}$

$P(X \leq x) = 1 - e^{-\lambda x}$

$\exp\left(\frac{1}{2}\right)$ $\lambda = 2$

$f(x) = \lambda e^{-\lambda x}$

$P(X \leq x)$ = death Prob

$P(X \geq x)$ = SERVIVAL prob

$$P(X \leq x) = \int_0^x \lambda e^{-\lambda x}\, dx = 1 - e^{-\lambda x}$$

$P(X \geq x) = 1 - (1 - e^{-\lambda x})$

SERVIVAL $= e^{-\lambda x}$

$$= \frac{e^{-1} \times \frac{1}{2}}{e^{-1} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2}} = \frac{\frac{1}{2e}}{\frac{1}{2e} + \frac{1}{4}} = \frac{\frac{1}{2e}}{\frac{2+e}{4e}}$$

$$= \frac{2}{2+e}$$

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$P(X \le x) = cdf = \int_0^x \lambda e^{-\lambda x} dx = \underline{1 - e^{-\lambda x}}$$

$$\frac{d}{dx}[F(x)] = f(x)$$

$$F(x) = \int_a^b f(x)\,dx$$

Exponential dist

- $P(X \le x)$ = Death prob = Len. $= 1 - e^{-\lambda x}$ — death
- $P(X \ge x) = 1 - (1 - e^{-\lambda x}) = e^{-\lambda x}$ — SERVIVAL
  SERVIVAL



Topic : Exponential distribution

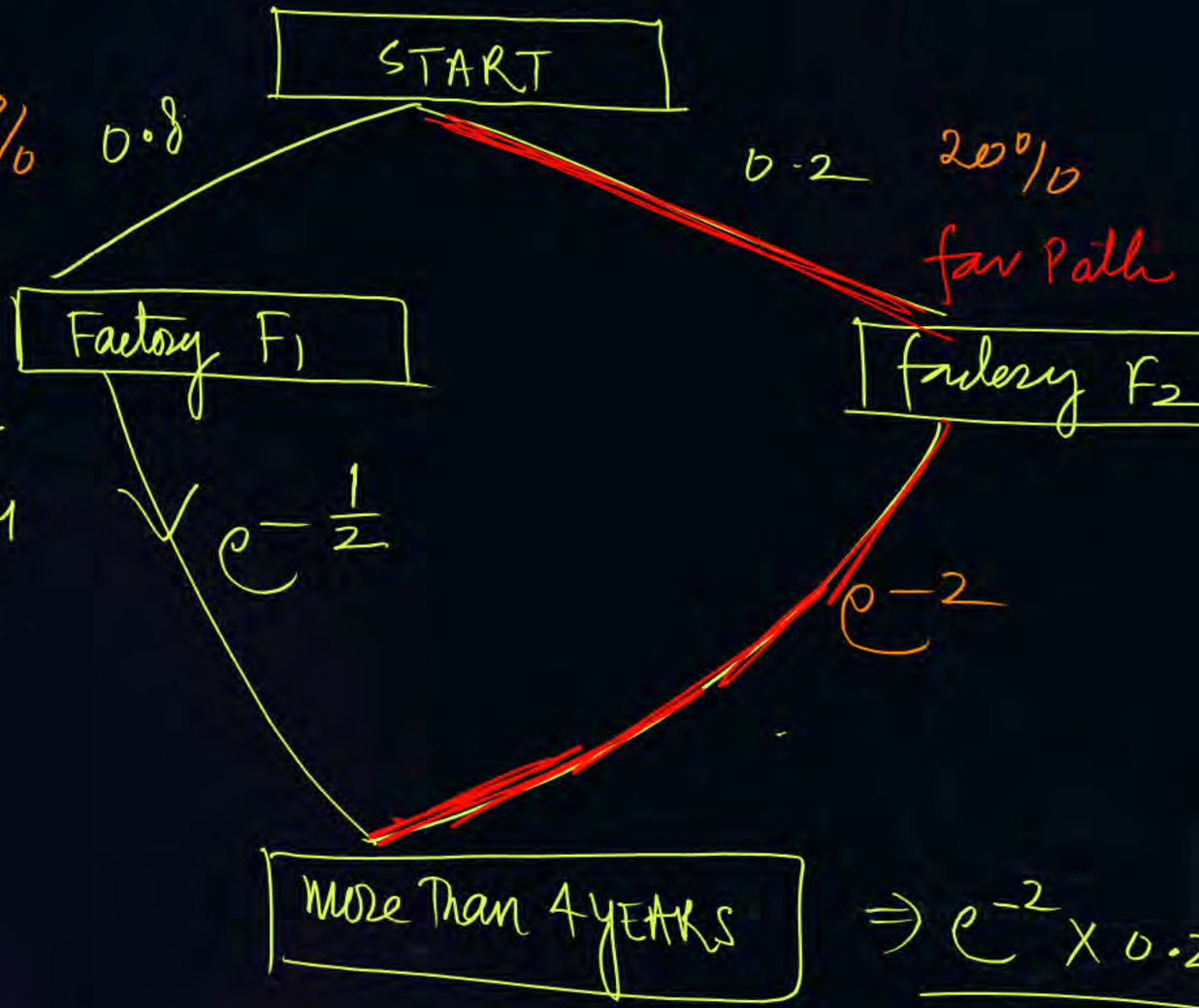
X_1 X_2
F_1 F_2
$\exp(\frac{1}{8})$ $\exp(\frac{1}{2})$

#Q. Let X1 and X2 denotes the lifetimes (in months) of bulbs produced at factories F1 and F2, respectively. Then random variable X1 and X2 are Exp(1/8) and Exp(1/2) respectively. A shop procures 80% of its supply of bulbs from factory F_1 and 20% from factory F_2. A randomly selected bulb from the shop is put on test and is found to be working after 4 months.

What is the probability that it was procured from factory F_2?

Markov Chain

Snake Ladder



$\exp\left(\frac{1}{8}\right)$

$P(X>4) = e^{-\lambda x}$

$= e^{-\frac{1}{8}\times 4}$

$= e^{-\frac{1}{2}}$

$\exp\left(\frac{1}{2}\right)$

$P(X>4)$

$= e^{-\lambda x}$

$= e^{-\frac{1}{2}\times 4}$

$= e^{-2}$

$$\Rightarrow \frac{e^{-2}\times 0.2}{e^{-2}\times 0.2 + 0.8\times e^{-\frac{1}{2}}} = 0.0528$$

Ans



Topic : Exponential distribution

at least = SERVIVAL
Atmost = death

#Q. Telephone call arrive at a switchboard following an exponential distribution with parameter $\lambda = 12$ per hour. If we are at the switchboard, what is the probability that the waiting time for a call is-

(i) At least 15 minutes

(ii) Not more than 10 minutes.

$\lambda = 12$ per Hours

$P(X \le x)$ = death $= 1 - e^{-\lambda x}$

$P(X \ge x) = e^{-\lambda x}$

$P\left(X \ge \frac{15}{60}\right) = P\left(X \ge \frac{1}{4}\right)$ = SERVIVAL

$P\left(X \le \frac{10}{60}\right) = P\left(X \le \frac{1}{6}\right)$ = death

$P\left(X \ge \frac{1}{4}\right) = e^{-12 \times \frac{1}{4}} = e^{-3}$

$P\left(X \le \frac{1}{6}\right) = 1 - e^{-12 \times \frac{1}{6}} = 1 - e^{-2}$



Topic : Exponential distribution

#Q. Show that for the exponential distribution:
$f(x) = Ae^{-x}$, $0 \leq x < \infty$, mean and variance are equal.

$f(x) = Ae^{-x} \quad (0 \leq x < \infty)$

mean $= 1/\lambda = 1$
variance $= 1/\lambda^2 = 1$

$A = 1$

$$\int_0^\infty Ae^{-x}\,dx = 1$$

$A = 1$ ✓


THANK - YOU